Abhay Bestrapalli

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EDUCATION

Massachusetts Institute of Technology (MIT) Cambridge, MA
B.S. in Computer Science and Engineering + Physics; GPA 4.8/5.0 May 2026
Relevant coursework: Deep Learning (G), Reinforcement Learning (G), Advances in Computer Vision, Accelerated Computing (G), Inference, Distributed Systems, Design and Analysis of Algorithms, Foundations of Information Policy

RESEARCH EXPERIENCE

Research Intern, Cartesia May 2025 – Aug 2025

- Built synthetic data to fix text-to-speech (TTS) pronunciation of acronyms and initialisms: curated ~3,500 terms, generated IPA pronunciations, synthesized 500K transcripts with Kokoro TTS on Kubernetes, and converted 200 hrs of it into diverse speaker voices with a voice-conversion model; audio-quality filters (MOS, silence, loudness, speaking rate) kept 87% of generated audio.
- Designed a 4-tier acronym eval including a held-out set of unseen acronyms; training on the data cut WER ~50% on the hard and out-of-distribution tiers (0.36 to 0.19 and 0.36 to 0.16) and held up as a small fraction of the full training mix.
- Built multilingual WER evals for TTS: benchmarked Gemini vs. Whisper as transcribers (Gemini stronger on multilingual, Whisper more reliable on English), wrote 20+ language-specific normalizers (CJK, Hindi, accents), and handled code-switched Hindi with per-word script transliteration.

Research Intern, Together AI May 2024 – Aug 2024

- Ran pretraining-data filtering ablations: trained 1.6B-parameter LLMs on 25–50B-token subsets of RedPajama-v2 selected by FineWeb-Edu classifier score (top 25% / 50%), evaluated on a ~30-task suite with DCLM-style centered accuracy.
- Found classifier quality scores uncorrelated with perplexity-based quality, indicating the two filters capture different signals; trained custom fastText quality classifiers using LLM-generated positives from Wikipedia.
- Built a synthetic supervised fine-tuning (SFT) data pipeline: generated ~24K document "scenarios" and combined them with Persona Hub personas to produce diverse instruction data.

Undergraduate Researcher, LIGO (Laser Interferometer Gravitational-Wave Observatory), MIT Spring 2023 – Spring 2024

- Applied ML classifiers to oLib, a low-latency, statistics-based detection pipeline for gravitational-wave transients, and evaluated its performance on injected (simulated) signals.

Undergraduate Researcher, TESS (Transiting Exoplanet Survey Satellite), MIT Fall 2022

- Automated quality flagging of exoplanet light curves in TESS's Quick Look Pipeline using random forest and decision tree classifiers.

SELECTED PROJECTS

LLM-Guided Reinforcement Learning | github.com/abhaybe/babyai-llm 6.7920 Deep RL, MIT, Fall 2025

- Developed two low-cost ways to bring LLMs into RL training: action suggestions only when the agent is stuck (<2% of steps), and one-shot LLM-generated reward-shaping code validated with AST checks and sandboxed execution.
- With PPO on sparse-reward MiniGrid tasks, raised success from 5% to 90% on DoorKey-8x8 and from 0% to 30% on MultiRoom-N6 at $0.05–0.10 of API cost per run; showed the methods are complementary (help-when-stuck alone fails on MultiRoom).

Scaling Kolmogorov-Arnold Networks for Vision (6.8301 Advances in CV, MIT, Spring 2024, team of 3): compared KAN scaling on CIFAR-10/100 by width vs. depth, mixture-of-experts, and MLP-Mixer architectures; width beat depth by up to 4 points at equal parameter count.

Geodesic Kaleidoscopes (HackMIT 2022) | github.com/abhaybe/geodesic-kaleidoscopes: web app that renders images by simulating light paths near black holes with general relativity.

TEACHING

Teaching Assistant, 6.7960 Deep Learning (graduate), MIT (Fall 2025): mentored student research projects; graded problem sets and projects and answered student questions.

Team Leader, USA Team, International Olympiad on Astronomy and Astrophysics (2023, Poland): moderated student papers, reviewed exam questions, and coached the team.

HONORS

- **International Olympiad on Astronomy and Astrophysics (IOAA):** Silver Medal 2021, Bronze Medal 2022 (USA team); **#1 in the USA**, National Astronomy Competition 2022
- **International Mathematical Olympiad Training Camp (India):** top 50 nationally, 2019 and 2020

SKILLS

ML: PyTorch, Hugging Face (Transformers, PEFT/LoRA), NumPy, pandas, scikit-learn, Weights & Biases, Ray, Daft, fastText
Languages & tools: Python, C/C++, Bash/Linux, Git, Docker, Kubernetes, LaTeX